



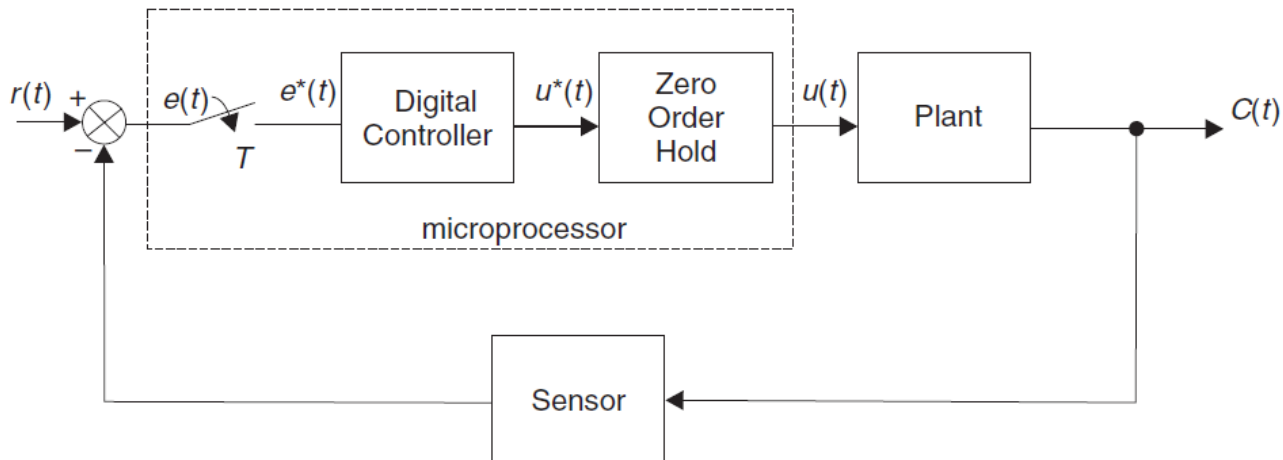
MCT5210 – Control Engineering III

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Digital Control Systems

- Developments in microprocessor technology have resulted in control algorithms being mainly implemented on microcontrollers.
- Before controllers can be implemented on microcontrollers, however, input **signals** must be **sampled** and converted into **digital** form.
- Microprocessor based control has the advantage of being versatile i.e. algorithms can easily be changed.
- It also enables implementation of sophisticated control laws including artificial intelligence based controllers.



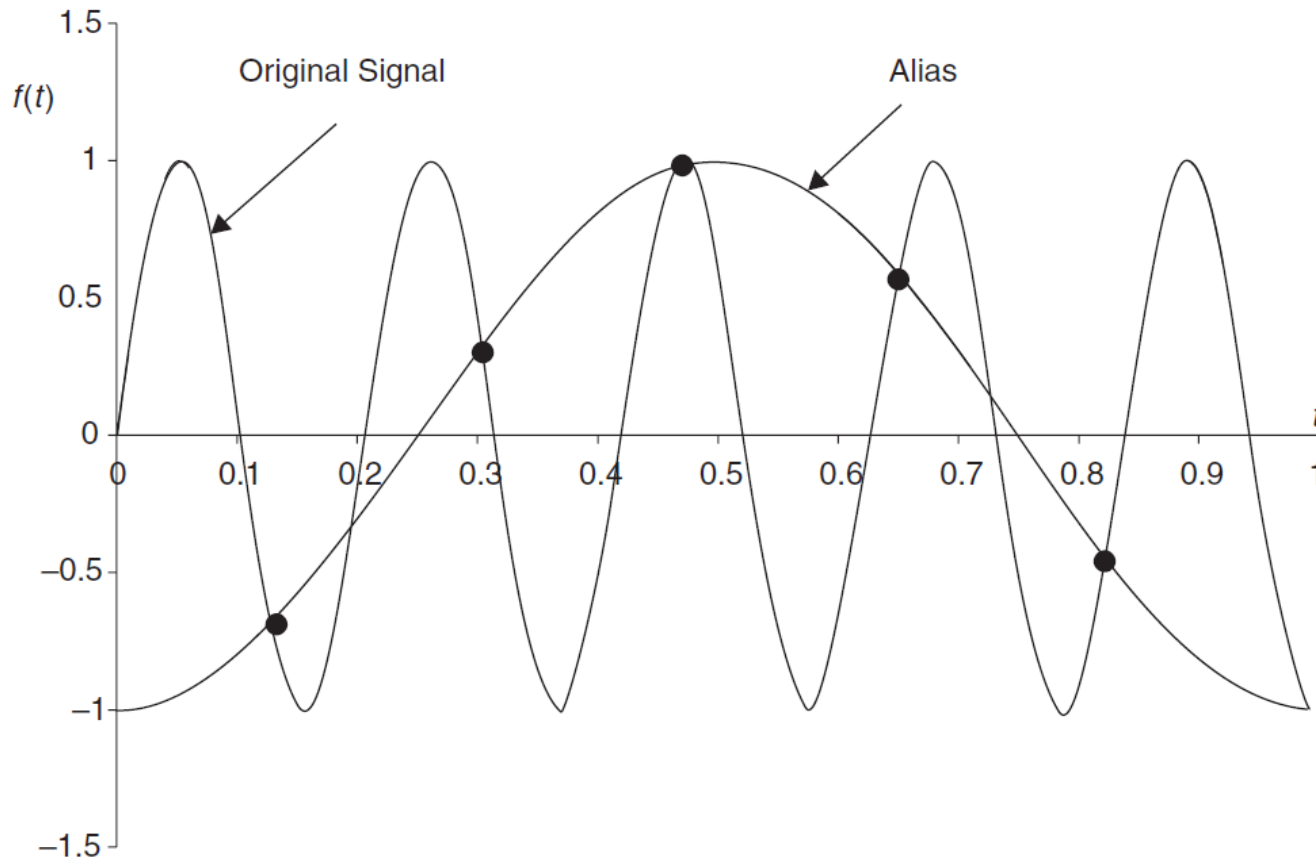


Sampled Data Systems

- The main **drawback** in **microprocessor** based control is that input data to the processor must be **sampled** via an A/D converter.
- Thus, only snap shots of the system data is available which makes the **sampling frequency** very important.
- **Shannon's sampling theorem** states that “A signal $f(t)$ that has a bandwidth ω_b is uniquely determined by a **discrete** set of sample values provided that the sampling frequency is **greater** than $2\omega_b$.
- The sampling frequency $2\omega_b$ is called the **Nyquist** frequency.
- A **lower frequency** signal called an **alias** may be constructed if the a signal is sampled below **Shannon's** limit.
- The rule of thumb is to sample a signal at about **ten times** higher than the highest frequency present in the signal.
- It is common practice to place an **anti aliasing filter**, which is an analogue low pass filter before the A/D converter to prevent aliasing.



Sampled Data Systems



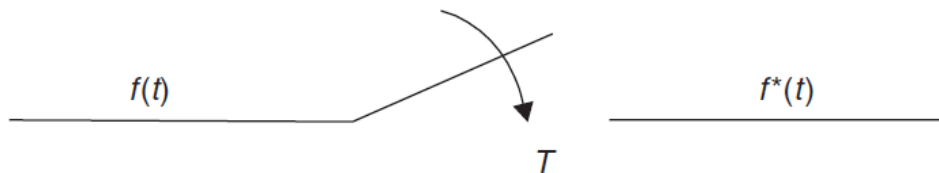


Sampled Data Systems

- In sampled data systems, is it assumed that all data enter or leave the microprocessor at the same fixed period T , called the **sampling period**.
- A sampler is basically a **switch** that **closes** every T seconds for one instant of time.
- An ideal sample $f^*(t)$ is a continuous signal $f(t)$ in a series of zero width impulses spaced at sampling time T seconds apart.
- The sampled signal can be represented by

$$f^*(t) = \sum_{k=0}^{\infty} f(kT) \delta(t - kT)$$

where $\delta(t - kT)$ is the **unit impulse** function occurring at $t = kT$.

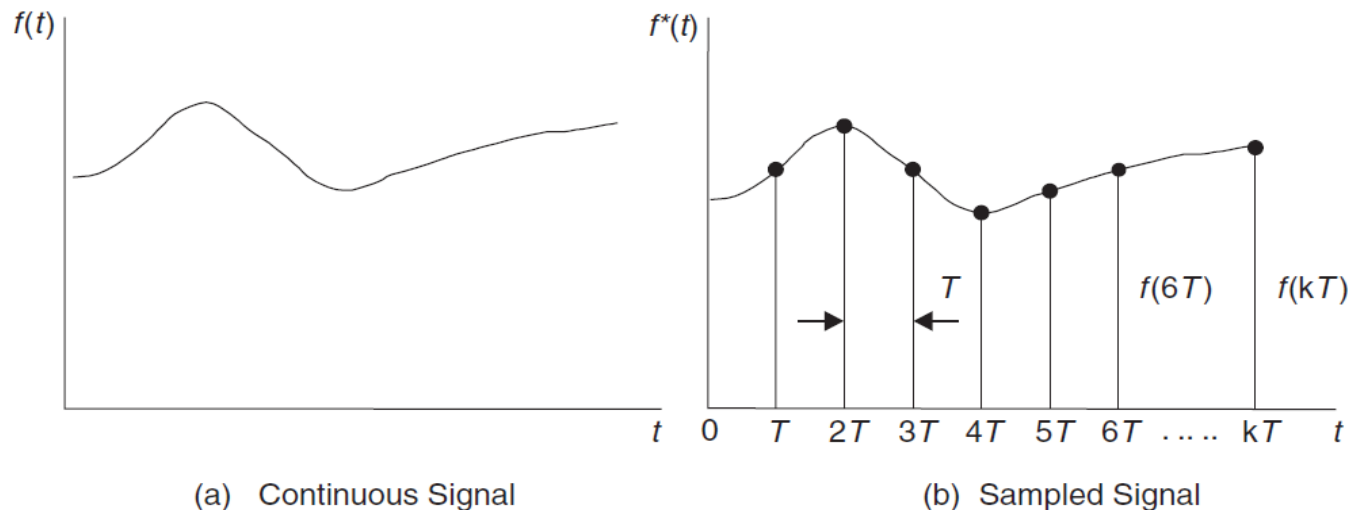




Sampled Data Systems

- It is possible to reconstruct $f(t)$ approximately from $f^*(t)$ by use of a **hold** device.
- The most common of this is a **zero order hold** which holds its output constant for one sample period thereby converting a series of **impulses** to a series of **pulses** of width T .
- The transfer function of a zero order hold is

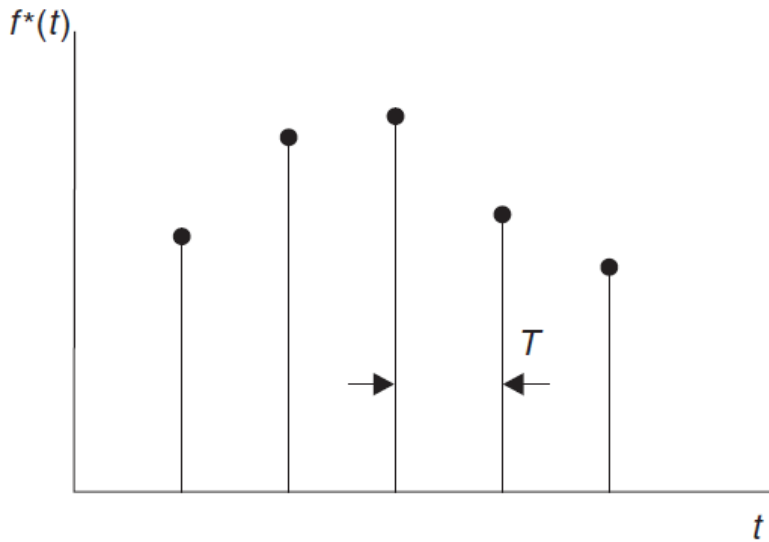
$$G_o(s) = \frac{1 - e^{-Ts}}{s}$$



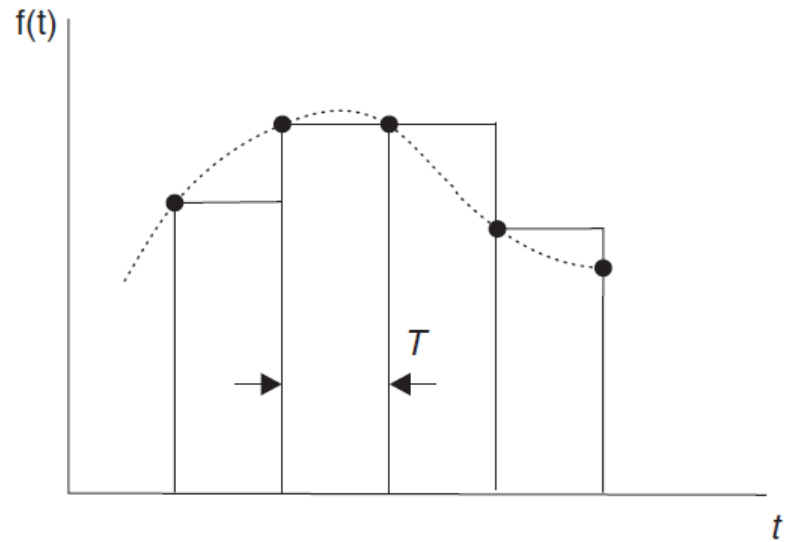
The sampling process



Sampled Data Systems



(a) Discrete Time Signal



(b) Continuous Time Signal

Construction of a continuous signal using a ZOH



The z-transform

- The **z-transform** is the main analytical tool for **discrete** time systems and is analogous to the Laplace transform for continuous systems.
- The symbol z is conceptually associated with discrete **time shifting** in a **difference equation** in the same manner s is associated with differentiation in a differential equation.

$$F^*(s) = \mathcal{L}[f^*(t)] = \sum_{k=0}^{\infty} f(kT) e^{-kTs}$$

$$F^*(s) = \sum_{k=0}^{\infty} f(kT) (e^{sT})^{-k}$$

Define z as

$$z = e^{sT}$$

Then

$$F(z) = \sum_{k=0}^{\infty} f(kT) z^{-k}$$

which can be written as

$$F(z) = f(0) + f(T)z^{-1} + f(2T)z^{-2} + \dots + f(kT)z^{-k}$$



The z-transform

Example: Find the z-transform of a unit step function $f(t) = 1$.

Define z as

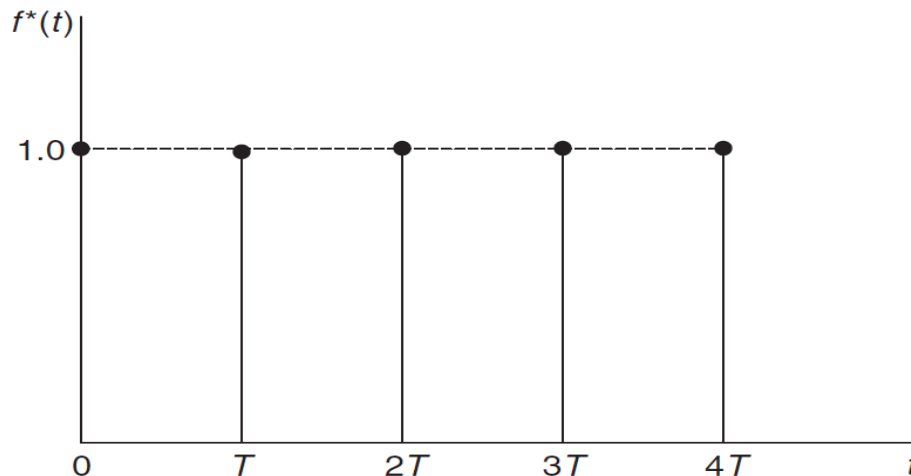
$$z[1(t)] = \sum_{k=0}^{\infty} 1(kT)z^{-k}$$

or

$$F(z) = 1 + z^{-1} + z^{-2} + \dots + z^{-k}$$

which in closed form is

$$z[1(t)] = \frac{z}{z-1} = \frac{1}{1-z^{-1}} \quad \rightarrow \text{Can be shown through long division}$$



$\rightarrow$ z-transform of a sampled unit step function



Common z-transforms

	$f(t)$ or $f(kT)$	$F(s)$	$F(z)$
1	$\delta(t)$	1	1
2	$\delta(t - kT)$	e^{-kTs}	z^{-k}
3	$1(t)$	$\frac{1}{s}$	$\frac{z}{z - 1}$
4	t	$\frac{1}{s^2}$	$\frac{Tz}{(z - 1)^2}$
5	e^{-at}	$\frac{1}{(s + a)}$	$\frac{z}{z - e^{-aT}}$
6	$1 - e^{-at}$	$\frac{a}{s(s + a)}$	$\frac{z(1 - e^{-aT})}{(z - 1)(z - e^{-aT})}$
7	$\frac{1}{a}(at - 1 + e^{-at})$	$\frac{a}{s^2(s + a)}$	$\frac{z\{(aT - 1 + e^{-aT})z + (1 - e^{-aT} - aTe^{-aT})\}}{a(z - 1)^2(z - e^{-aT})}$
8	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	$\frac{z \sin \omega T}{z^2 - 2z \cos \omega T + 1}$
9	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$	$\frac{z(z - \cos \omega T)}{z^2 - 2z \cos \omega T + 1}$
10	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$	$\frac{ze^{-aT} \sin \omega T}{z^2 - 2ze^{-aT} \cos \omega T + e^{-2aT}}$
11	$e^{-at} \cos \omega t$	$\frac{(s + a)}{(s + a)^2 + \omega^2}$	$\frac{z^2 - ze^{-aT} \cos \omega T}{z^2 - 2ze^{-aT} \cos \omega T + e^{-2aT}}$



Inverse Transformation

Example 1: A sampled data system has the transfer function

$$G(s) = \frac{1}{s + 1}$$

If the sampling time is 1 s and the system is subject to a unit step input, determine the discrete time response using a zero order hold.

Inverse Transformation: Difference Equation

Alternatively, consider a system of the form

$$\frac{Y(z)}{U(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots}$$

Thus

$$Y(z) = (-a_1 z^{-1} - a_2 z^{-2} - \dots)Y(z) + (b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots)U(z)$$

Which can be expressed as a **difference equation** in the form

$$y(kT) = -a_1 y(k-1)T - a_2 y(k-2)T - \dots \\ + b_0 u(kT) + b_1 u(k-1)T + b_2 u(k-2)T + \dots$$

$$\frac{Y(z)}{U(z)} = \frac{0.632}{z - 0.368} = \frac{0.632z^{-1}}{1 - 0.368z^{-1}}$$

$$Y(z) = 0.632z^{-1}Y(z) + 0.368z^{-1}U(z)$$

Thus

$$y(kT) = 0.368y(k-1)T + 0.632u(k-1)T$$

$$y(0) = 0.368y(-1) + 0.632u(-1) = 0 + 0 = 0, \quad k = 0$$

$$y(1) = 0.368y(0) + 0.632u(0) = 0 + 0.632 = 0.632, \quad k = 1$$

$$y(2) = 0.368y(1) + 0.632u(1) = 0.233 + 0.632 = 0.865, \quad k = 2$$